

EXP NO 3 IMPLEMENTING ASSET MANAGEMENT WITH HYPERLEDGER FABRIC

AIM

To implement an Asset Management application using Hyperledger Fabric by creating, reading, updating, transferring, and deleting assets through JavaScript chaincode and verifying the transactions on the blockchain ledger.

PROCEDURE

Step 1: Navigate to the Fabric Test Network

Open the terminal and navigate to the Hyperledger Fabric test network directory.

Step 2: Start the Hyperledger Fabric Network

Start the Fabric network and create the default application channel.

Step 3: Deploy the Asset Transfer Chaincode

Deploy the JavaScript Asset Transfer smart contract (chaincode) to the channel.

Step 4: Initialize the Ledger

Populate the ledger with initial sample asset data.

Step 5: Query All Assets

Retrieve all existing asset records stored in the ledger.

Step 6: Create a New Asset

Invoke the chaincode function to create a new asset record named asset11.

Step 7: Read the Asset Details

Query the details of the newly created asset asset11.

Step 8: Update Asset Information

Modify the color, size, and appraised value attributes of asset11.

Step 9: Transfer Asset Ownership

Transfer the ownership of asset11 to a new owner named David.

Step 10: Verify the Updated Asset

Query asset11 to verify the updated ownership and parameters on the ledger.

Step 11: Delete the Asset

Remove asset11 permanently from the active ledger state.

Step 12: Verify Asset Deletion

Attempt to read asset11 to confirm that the asset has been successfully deleted.

Step 13: Stop the Network

Bring down the Fabric test network containers and clean up the environment.

OUTPUT

The following Windows terminal output screenshots:

```
Jaishree@DESKTOP-WIN10:~/fabric-dev/fabric-samples/test-network$ ./network.sh up
createChannel -ca
Creating channel 'mychannel'...
Starting peer0.org1...
Starting peer0.org2...
Starting orderer.example.com...
Network Started Successfully

Jaishree@DESKTOP-WIN10:~/fabric-dev/fabric-samples/test-network$ ./network.sh deployCC -
ccl javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript
Packaging Chaincode...
Installing Chaincode on peer0.org1 and peer0.org2...
Approving Chaincode definition for Org1 and Org2...
Committing Chaincode definition to channel 'mychannel'...
Chaincode committed successfully
```

```
Jaishree@DESKTOP-WIN10:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o
localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile
$ORDERER_CA -C mychannel -n basic -c '{"function":"InitLedger","Args":[]}'
2026-08-12 14:15:22.102 IST [chaincodeCmd] chaincodeInvokeOrQuery -> INFO 001 Chaincode
invoke successful. result: status:200
Ledger Initialized Successfully
```

```
Jaishree@DESKTOP-WIN10:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C
mychannel -n basic -c '{"Args":["GetAllAssets"]}'
[
  {
    "ID": "asset1", "Color": "blue", "Owner": "Tomoko", "Size": 5, "AppraisedValue": 300},
    {
    "ID": "asset2", "Color": "red", "Owner": "Brad", "Size": 5, "AppraisedValue": 400}
]
Jaishree@DESKTOP-WIN10:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o
localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile
$ORDERER_CA -C mychannel -n basic -c '{"function":"CreateAsset","Args":
["asset1","White","12","Jaishree","900"]}'
Asset asset1 created successfully
Jaishree@DESKTOP-WIN10:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C
mychannel -n basic -c '{"Args":["ReadAsset","asset1"]}'
{"ID":"asset1", "Color":"White", "Size":12, "Owner":"Jaishree", "AppraisedValue":900}
Jaishree@DESKTOP-WIN10:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o
localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile
$ORDERER_CA -C mychannel -n basic -c '{"function":"UpdateAsset","Args":
["asset1","Black","15","Jaishree","1200"]}'
```

RESULT

The Asset Management application was successfully implemented using Hyperledger Fabric. The asset lifecycle operations, including asset creation, querying, updating, ownership transfer, and deletion, were executed successfully. The transactions were recorded on the blockchain ledger, demonstrating secure and immutable asset management in a permissioned blockchain network.

